

CS 5785/ECE 5414/ORIE 5750 Applied Machine Learning

## Lecture 2

*Supervised Learning + Linear Regression*

# Announcements

- Complete two surveys on Canvas
- Homeworks 0 and 1 are out on Canvas
  - HW0 Due on 9/2 (Wed)
  - HW1 Due on 9/16 (Wed)
- Office hours:
  - Kyra: Monday 5- 6:30 pm Bloomberg 464
    - Next week: on Wednesday 5-6:30 pm (other exception dates listed on Canvas)
  - Adnan: Thursday 10-11:30 am
  - Jindan: Tuesday 5-6:30 pm
  - Diyang: Thursday 4-5:30 pm

# Announcements

- Review sessions:
  - Recording and slides have been posted on canvas

## Recall: Machine Learning Basics

- Input data
- Output (targets task, target values)
- Model class
- Loss function/objective function
- Learning rule/optimizer

# Data Representation: Predicting Diabetes

|   | age       | sex       | bmi       | bp        | s1        | s2        | s3        | s4        | s5        | s6        |
|---|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| 0 | 0.038076  | 0.050680  | 0.061696  | 0.021872  | -0.044223 | -0.034821 | -0.043401 | -0.002592 | 0.019908  | -0.017646 |
| 1 | -0.001882 | -0.044642 | -0.051474 | -0.026328 | -0.008449 | -0.019163 | 0.074412  | -0.039493 | -0.068330 | -0.092204 |
| 2 | 0.085299  | 0.050680  | 0.044451  | -0.005671 | -0.045599 | -0.034194 | -0.032356 | -0.002592 | 0.002864  | -0.025930 |
| 3 | -0.089063 | -0.044642 | -0.011595 | -0.036656 | 0.012191  | 0.024991  | -0.036038 | 0.034309  | 0.022692  | -0.009362 |
| 4 | 0.005383  | -0.044642 | -0.036385 | 0.021872  | 0.003935  | 0.015596  | 0.008142  | -0.002592 | -0.031991 | -0.046641 |

- bmi body mass index
- bp average blood pressure
- s1 tc, T-Cells (a type of white blood cells)
- s2 ldl, low-density lipoproteins
- s3 hdl, high-density lipoproteins
- s4 tch, thyroid stimulating hormone
- s5 ltg, lamotrigine
- s6 glu, blood sugar level

# Data Representation: Predicting Diabetes

|   | age       | sex       | bmi       | bp        | s1        | s2        | s3        | s4        | s5        | s6        |
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- Predict whether a patient has diabetes based on vectors of patient attributes.
- Vector representation of the data  $x = [x_1, x_2, x_3, \dots, x_{10}]$ 
  - $x_1$ : age
  - $x_2$ : sex
  - $x_3$ : BMI
  - $x_4$ : average blood pressure (bp)
  - ...

# Data Representation: Predicting Diabetes

|   | age       | sex       | bmi       | bp        | s1        | s2        | s3        | s4        | s5        | s6        |
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```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
plt.rcParams['figure.figsize'] = [12, 4]
from sklearn import datasets

# Load the diabetes dataset
diabetes = datasets.load_diabetes(as_frame=True)
print(diabetes.DESCR)
```

```
# Load the diabetes dataset
diabetes_X, diabetes_y = diabetes.data, diabetes.target

# Print part of the dataset
diabetes_X.head()
```

# Data Representation: Predicting Diabetes

- The amount of signal that you will be able to extract highly depends on the preprocessing steps
- Discrete variables (ordering matters, e.g., age)
- Continuous variables (e.g., lab values)
- Categorical variables (e.g., zipcode)
  - E.g., Data column containing
    - 10040, 10041, 10042, 10043, 10044
  - How do we represent this?

## How to represent the 5 zipcodes in the example?



convert it to numerical values and use it as it is

0

convert it to numerical values and standardize it

0

represent it using a binary vector of length 5, where  
each entry represents one zipcode

0

Any of the above works well in practice

0

# Data Representation: Predicting Diabetes

- The amount of signal that you will be able to extract highly depends on the preprocessing steps
- Discrete variables (ordering matters, e.g., age)
- Continuous variables (e.g., lab values)
- Categorical variables (e.g., zipcode)
  - E.g., Data column containing
    - 10040, 10041, 10042, 10043, 10044
  - How do we represent this?
  - One-hot encoding
    - Use a vector of length 5 to represent zipcode
    - E.g., the value 10044 is replaced with [0,0,0,0,1]

# Data Representation: Predicting Diabetes

- When do we want to **convert** continuous values to discrete or categorical variables
- E.g., when different lab has different ways of processing ranges
  - Labs use different amounts of blood/technology to count the number of cells
  - Whether your red blood cell count is normal depends on how your blood is physically processed in the lab
  - Might be more meaningful to convert blood cell counts to categorical values: normal, abnormal, low flag, high flag, and so on.

# Data Representation: Predicting Diabetes

- Create an additional feature indicating whether the patient is old and a man
- Useful if old men are at risk

```
diabetes_X['old_man'] = (diabetes_X['sex'] > 0) & (diabetes_X['age'] > 0.05)
diabetes_X.head()
```

|   | age       | sex       | bmi       | bp        | s1        | s2        | s3        | s4        | s5        | s6        | old_man |
|---|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|---------|
| 0 | 0.038076  | 0.050680  | 0.061696  | 0.021872  | -0.044223 | -0.034821 | -0.043401 | -0.002592 | 0.019908  | -0.017646 | False   |
| 1 | -0.001882 | -0.044642 | -0.051474 | -0.026328 | -0.008449 | -0.019163 | 0.074412  | -0.039493 | -0.068330 | -0.092204 | False   |
| 2 | 0.085299  | 0.050680  | 0.044451  | -0.005671 | -0.045599 | -0.034194 | -0.032356 | -0.002592 | 0.002864  | -0.025930 | True    |
| 3 | -0.089063 | -0.044642 | -0.011595 | -0.036656 | 0.012191  | 0.024991  | -0.036038 | 0.034309  | 0.022692  | -0.009362 | False   |
| 4 | 0.005383  | -0.044642 | -0.036385 | 0.021872  | 0.003935  | 0.015596  | 0.008142  | -0.002592 | -0.031991 | -0.046641 | False   |

## Is it beneficial to maximize the number of input features in a machine learning model?

It depends on the model class, for more complex models, the more the better

0

It depends on the size of the dataset, for large datasets, the more the better

0

It is always beneficial

0

It is never beneficial

0

None of the above

0

## Dataset: Attributes

- Formally, we call columns in the dataset as attributes
  - $i$ : the  $i$ th patient in the dataset
  - An attribute  $x^{(i)} \in \mathcal{X}$  is a  $d$ -dimensional vector of the form

$$x^{(i)} = \begin{bmatrix} x_1^{(i)} \\ x_2^{(i)} \\ \dots \\ x_d^{(i)} \end{bmatrix}$$

- The set  $\mathcal{X}$ : attribute space
  - Intuitively: contains all possible values that attributes can take
  - Oftentimes:  $\mathcal{X} = \mathbb{R}^d$

## Dataset: Features

- Features: representations of the attributes
- Function  $\phi: \mathcal{X} \rightarrow \mathbb{R}^p$  inputs  $x^{(i)} \in \mathcal{X}$  and outputs a p-dimensional representation of the data

$$\phi(x^{(i)}) = \begin{bmatrix} \phi(x_1^{(i)}) \\ \phi(x_2^{(i)}) \\ \dots \\ \phi(x_d^{(i)}) \end{bmatrix}$$

- $\phi(x^{(i)})$ : featurized input,  $\phi(x^{(i)})_j$ : a feature.
- Attributes and features are often used interchangeably
  - Most authors refer  $x^{(i)}$  as features
  - Will follow this convention and use attributes when there is ambiguity

# Target Values/Labels

Target labels decide the supervised learning task:

- Regression
  - Predicting a **variable** that is continuous or (a lot of) discrete values
    - dollar in sales, # of days since last hospital admission
- Binary classification
  - Predicting a **variable** that takes binary values
    - Email spam/not, patient has diabetes/not

# Target Values/Labels

- Multiclass classification
  - Predicting a **categorical variable** that takes  $\geq 3$  values
    - Which one of 10 digits applies to this image?
- Multilabel classification
  - Categories are not mutually exclusive
    - Topics of a news article
- Structured prediction
  - Produce a complex object like a sentence, image, graph, or molecule
    - Machine translation

## Dataset: Target Values / Labels

- Formally, the target variable is represented by  $y^{(i)} \in \mathcal{Y}$ 
  - $i$ : the  $i$ th patient in the dataset
  - The set  $\mathcal{Y}$ : target space
    - Contains all possible values that the target can take
    - Assume  $y^{(i)}$  is a one-dimensional vector from now on

## A Supervised Learning Dataset: Notation

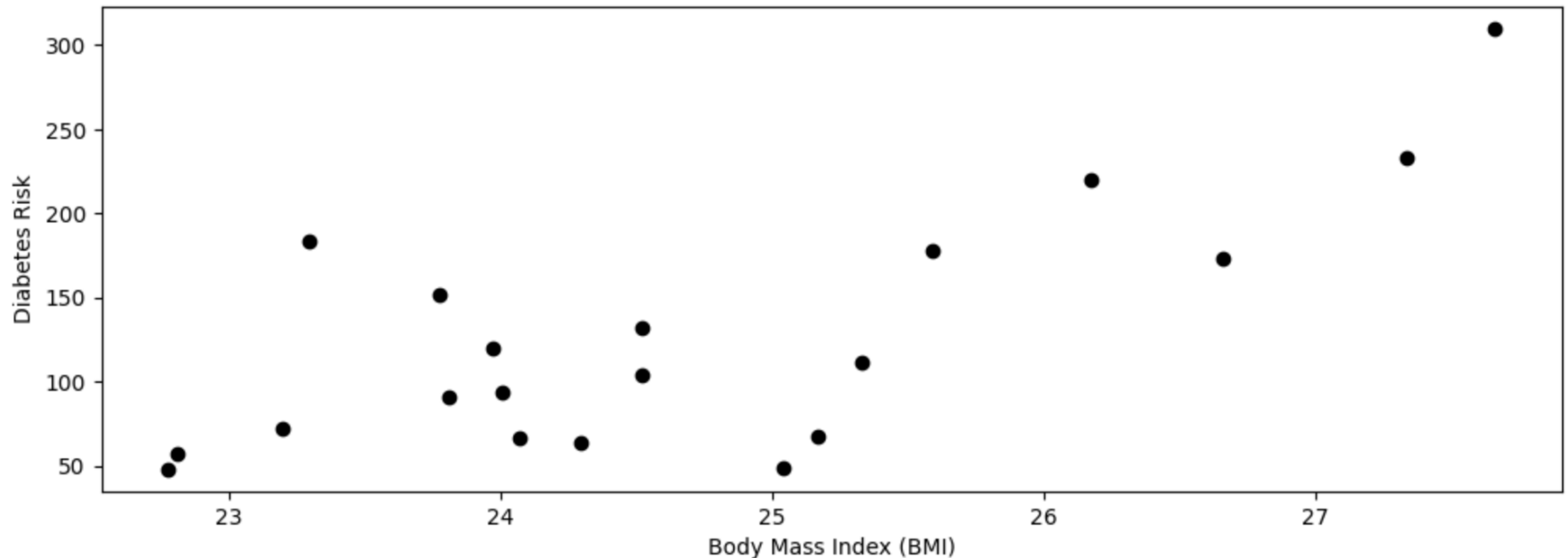
- Formally, the training dataset of size  $n$  (e.g., containing  $n$  patients) is represented by a set  $\mathcal{D}$

$$\mathcal{D} = \{(x^{(i)}, y^{(i)}) | i = 1, 2, \dots, n\}$$

- $x^{(i)} \in \mathcal{X}$ : features/inputs (e.g., measurements collected from patient  $i$ )
- $y^{(i)} \in \mathcal{Y}$ : target (e.g., diabetes risk)
- $(x^{(i)}, y^{(i)})$ : training example

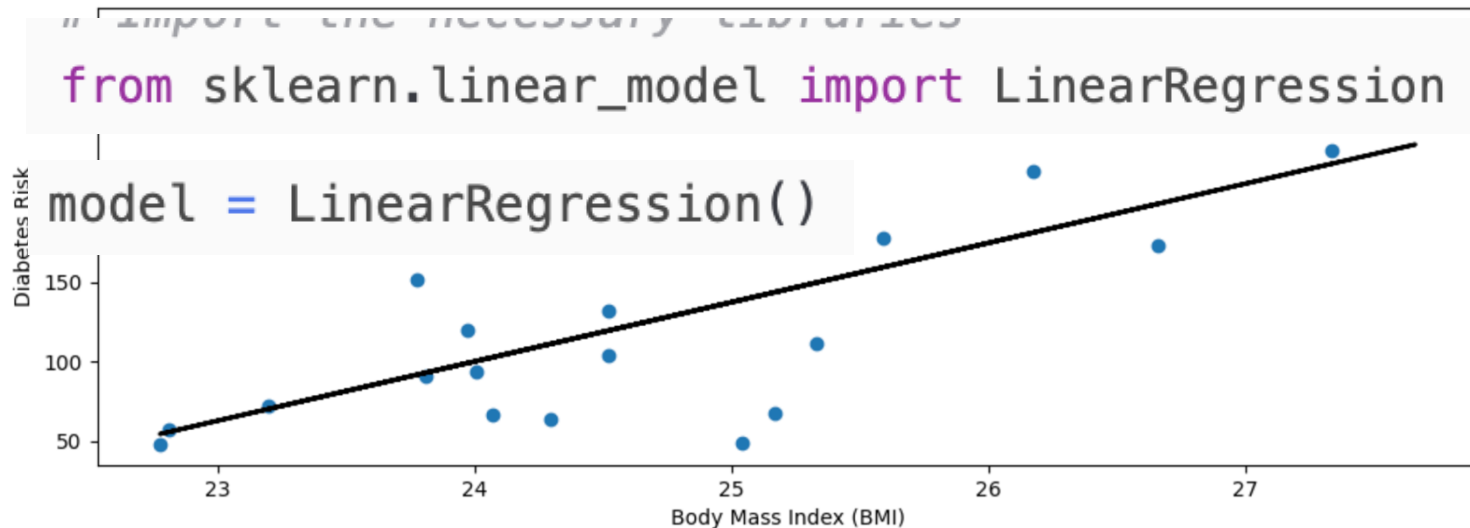
## Example: A Supervised Learning Dataset

- Predicting diabetes risk
- For each patient we are interested in predicting diabetes risk using BMI



## Example: A supervised Learning Dataset

- How do we explain the correlation between the feature BMI and label diabetes risk?
- Data looks linear!
- Might be able to fit a line with slope to explain the variability in the data
- Linear regression! → this is a model class that we decided to pick



## Model: Notation

- Model: a function that maps inputs  $x \in \mathcal{X}$  to targets  $y \in \mathcal{Y}$ :

$$f: \mathcal{X} \rightarrow \mathcal{Y}$$

- Linear regression (predicting diabetes risk using BMI):

$$f(x) = \theta_0 + \theta_1 x$$

for some  $\theta_0, \theta_1 \in \mathbb{R}$ , e.g.,  $\theta_0 = 0, \theta_1 = 1$

- Often, models are parametrized by parameters  $\boldsymbol{\theta} \in \Theta$ 
  - we write the following to denote parametric models

$$f_{\boldsymbol{\theta}}: \mathcal{X} \rightarrow \mathcal{Y}$$

- In the above linear regression  $\boldsymbol{\theta} = [\theta_0, \theta_1]$

## Model Class: Notation

- A model class is a **set** of possible models that map input features to targets:

$$\mathcal{M} \subseteq \{f \mid f: \mathcal{X} \rightarrow \mathcal{Y}\}$$

- Model class of the running linear regression example

$$\mathcal{M} = \{f_{\theta}(x) \mid f_{\theta}(x) = \theta_0 + \theta_1 x : \theta_0, \theta_1 \in \mathbb{R}\}$$

- Objective of learning: selecting the model with the **best fit** within the model class

# Choosing a Model Class

Must restrict hypothesis space to generalize

- Otherwise, we can just overfit
  - Without assumptions, my current data is not informative of unseen future data
- Does not mean that my hypothesis space has to be finite
- Linear regression
  - The number of potential values that the parameters can take is infinite: the coefficients are continuous
- Each model class satisfy different properties

# Properties of Linear Regression

- **Independence**: each feature exerts an independent effect on the prediction
- **Monotonicity**: increasing feature  $x_i$  makes  $f(\mathbf{x})$  go **up always** or **down always**
  - This assumption does not hold in general (Age and income)
- **Uniform effects**
  - Increasing a feature by some fixed amount  $\mathbf{x} \rightarrow \mathbf{x} + \Delta_1$  should influence the model's output by a corresponding amount  $f(\mathbf{x} + \Delta_1) \rightarrow f(\mathbf{x}) + \Delta_2$
  - Assumption violation (income and happiness)

# Choosing a Model Class

There are many families of models

- For many tasks today, neural networks dominate
  - High-dimensional data
  - Large amount of data
  - Function fitting
- They haven't always, and may not in the future
- Many classical methods (e.g., linear models) are special cases of neural networks, so they are intuitive to learn

# Objective functions

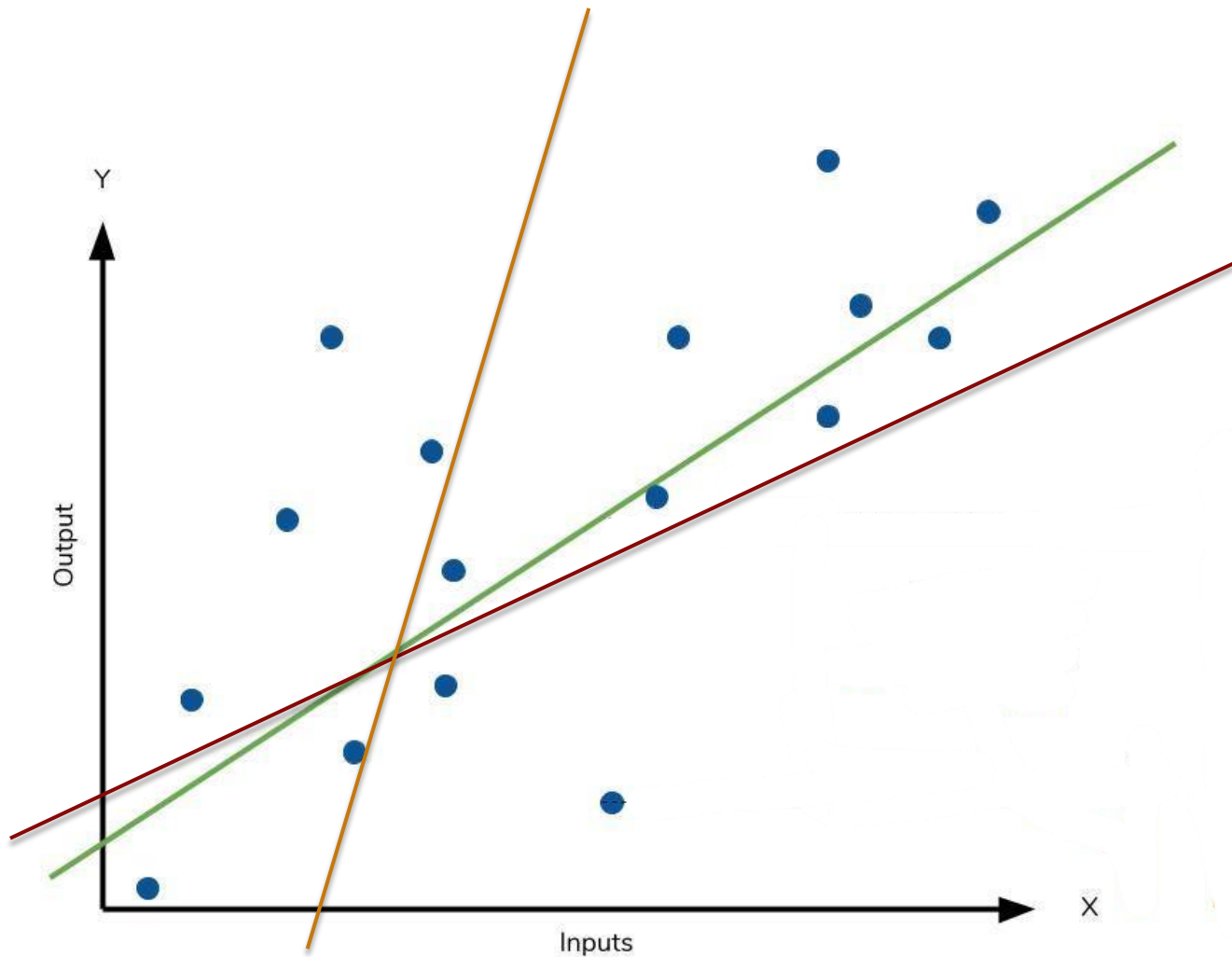
- What do we mean by **best fit**?
- Supervised learning: fit is often measured by the distance between the predicted outcome and the true label
- Aka, the use of loss functions

$$loss_{\theta}(\hat{y}^{(i)}, y^{(i)})$$

- The way that we measure the distance between two vectors are not unique
- There are often multiple ways of selecting an objective function

# Objective functions

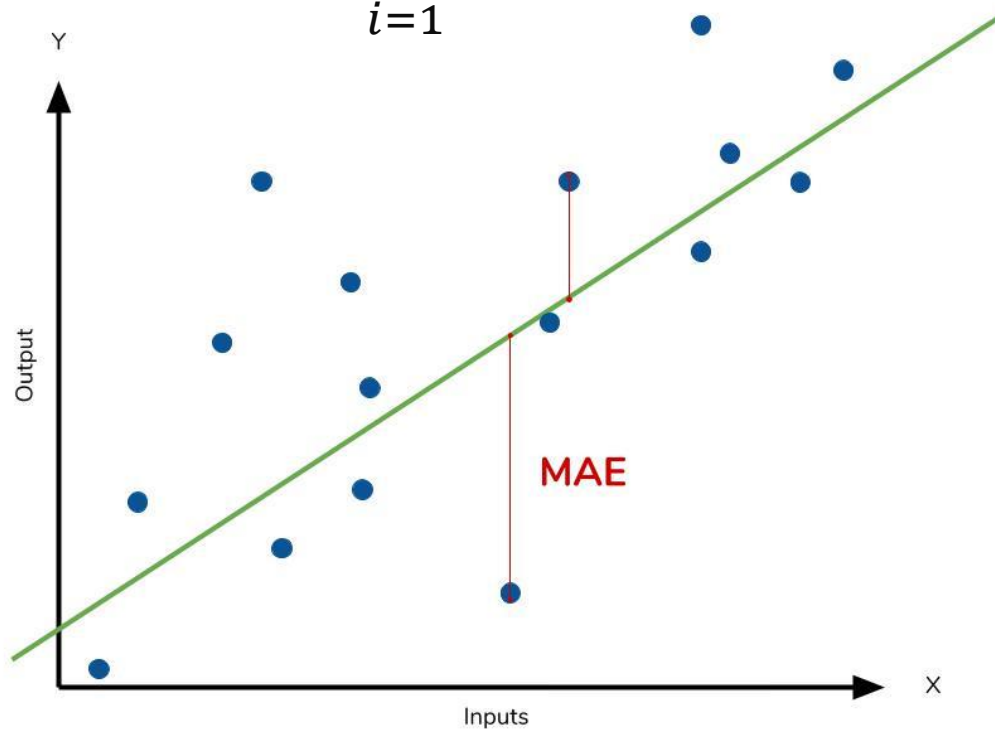
- Regression



# Objective functions

- Regression
- Absolute error

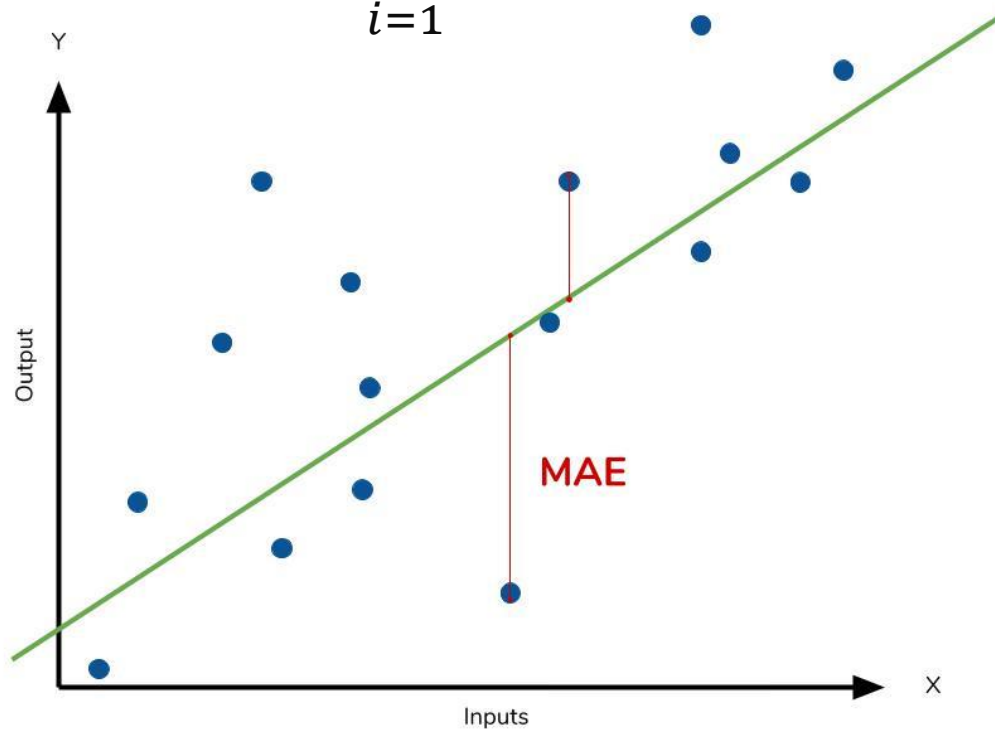
$$\text{loss}_{\theta} = \frac{1}{n} \sum_{i=1}^n |f_{\theta}(x^{(i)}) - y^{(i)}|$$



# Objective functions

- Regression
- Absolute error

$$\text{loss}_{\theta} = \frac{1}{n} \sum_{i=1}^n |f_{\theta}(x^{(i)}) - y^{(i)}|$$



# Objective functions

- Regression
  - Mean squared error (most common in linear regression)

$$\text{loss}_{\theta} = \frac{1}{n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2$$

